



EAST WEST UNIVERSITY

TASK-1 REPORT

EuroSAT: Land Use and Land Cover Classification

Course Code: CSE475
Course Title: Machine Learning
Section: 06
Semester: Summer 2026

Group – 5

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Related Work

This section reviews five studies closely related to land use and land cover classification on the EuroSAT dataset, summarizing each study's methodology, reported performance, limitations, strengths, and relevance to this project.

1. Interpretable Deep Learning Framework for LULC Classification in Remote Sensing Using SHAP

Authors	Temenos et al.
Year	2023
Dataset	EuroSAT
Application Area	Land use / land cover classification with band-importance analysis
Method	Compact CNN + SHAP (deep explainer)
Metrics	Accuracy: 94.72%

- **Strengths:** Provides explainable band selection; reduces input data by 77% while keeping accuracy high; improves per-class accuracy.
- **Limitations:** Accuracy is capped below other CNN benchmarks (~98%); SHAP explanation is computationally expensive.
- **Research Gap:** Most EuroSAT studies report only accuracy without explaining which spectral bands actually drive the predictions.
- **Relation to This Work:** Supports this project's use of explainability/interpretability alongside CNN classification on the EuroSAT dataset.

2. Kolmogorov–Arnold Network for Satellite Image Classification in Remote Sensing

Authors	Cheon, M.
Year	2024
Dataset	EuroSAT – Sentinel-2 satellite images, 10 LULC classes
Application Area	Satellite / land-use image classification exploring a novel MLP-alternative architecture
Method	Kolmogorov–Arnold Network (KAN), benchmarked against ConvNeXt, ResNet101, VGG16, MobileNetV2, EfficientNet, ViT

Metrics	Best result: ConvNeXt + KAN reaching 96% accuracy (starting at 94% at epoch 1)
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- **Strengths:** Introduces a newer, interpretable alternative to standard MLP classifier heads; benchmarks against several established CNN/ViT backbones.
- **Limitations:** Only preprint-level accuracy (96%), below top CNN/ViT benchmarks (~98%+); KAN training is less mature/optimized than standard architectures.
- **Research Gap:** KAN is still new and under-optimized for remote sensing, so its accuracy has not yet matched top CNN/ViT benchmarks (~98%+).
- **Relation to This Work:** Shows an alternative architecture that can be referenced or compared against when exploring beyond standard CNNs.

3.Deep Learning-Based Spatial Pattern Modeling for Land Use and Land Cover Classification Using Satellite Imagery

Authors	Marzougui, M. et al.
Year	2025
Dataset	EuroSAT – 27,000 Sentinel-2 images, 10 LULC classes
Application Area	Spatial pattern modeling for LULC classification, environmental monitoring
Method	InceptionV3 (fine-tuned hyperparameters)
Metrics	Accuracy: 99.43%; Validation accuracy: 98.50%; Training loss: 0.0184; Validation loss: 0.0566

- **Strengths:** Very high accuracy (99.43%); reports both training and validation metrics, showing the model is not overfitting.
- **Limitations:** Single-architecture study (only tests InceptionV3, no comparison against other CNNs); published in a meteorology-focused journal rather than a CV/remote-sensing venue.
- **Research Gap:** Only tests a single architecture (InceptionV3), so it does not compare performance against other CNN models on EuroSAT.
- **Relation to This Work:** Gives a strong accuracy benchmark to compare this project's model performance against on the same EuroSAT dataset.

4.Mapping of Land Use and Land Cover (LULC) using EuroSAT and Transfer Learning

Authors	Kunwar, S., Ferdush, J.
Year	2023/2024
Dataset	EuroSAT (RGB) – 27,000 images, 64×64 px, 10 classes
Application Area	LULC mapping applied to a real region (Kreis Borken, Germany)

Method	Vision Transformer (ViT) pre-trained on ImageNet-21k, fine-tuned via transfer learning
Metrics	Accuracy: 99.19%

- **Strengths:** Highest accuracy among reviewed work (99.19%); demonstrates a real-world mapping use case beyond simple classification.
- **Limitations:** Uses only RGB bands; transformer training needs GPU resources.
- **Research Gap:** Only uses RGB bands and ignores the other 10 Sentinel-2 spectral bands, so multispectral information is left unused.
- **Relation to This Work:** Gives a transfer-learning / transformer baseline to compare against if this project uses CNNs on the same RGB EuroSAT data.

5. Benchmarking and Scaling of Deep Learning Models for Land Cover Image Classification

Authors	Papoutsis et al.
Year	2023
Dataset	BigEarthNet (Sentinel-2, multi-label) – further evaluated on EuroSAT and SEN12MS
Application Area	Large-scale LULC benchmarking and transfer learning
Method	CNN zoo (60+ models); EfficientNet-scaled Wide ResNet + Efficient Channel Attention
Metrics	+4.5% F1-score over ResNet-50 baseline (BigEarthNet, 19 classes)

- **Strengths:** Extensive model zoo (60+ models); domain-relevant pre-trained weights improve transfer performance to EuroSAT.
- **Limitations:** EuroSAT is a secondary evaluation, not the primary focus; the large-scale setup is compute-heavy.
- **Research Gap:** EuroSAT is only a secondary evaluation dataset here, so there isn't deep EuroSAT-specific analysis.
- **Relation to This Work:** Suggests that pre-training on a larger remote-sensing dataset before fine-tuning could improve this project's accuracy on EuroSAT.

Exploratory Data Analysis of the EuroSAT Dataset

Introduction

Exploratory Data Analysis (EDA) is an important step before developing any machine learning or deep learning model. It helps understand the dataset, identify missing values, analyze data

distributions, detect class imbalance, and gain insights that guide preprocessing and model development.

Dataset Description

The EuroSAT dataset contains 27,000 labeled and geo-referenced satellite images covering land use and land cover categories, based on Sentinel-2 satellite imagery.

Total Images	27,000
Number of Classes	10
Source Satellite	Sentinel-2 (ESA Copernicus program)
Image Format	JPEG (RGB version) / TIFF (multispectral version)
Image Type	RGB (3 bands) or Multispectral (13 bands)
Image Size	64 × 64 pixels
Spatial Resolution	10 m per pixel

Class Labels

- Annual Crop
- Forest
- Herbaceous Vegetation
- Highway
- Industrial
- Pasture
- Permanent Crop
- Residential
- River
- Sea/Lake

Objectives of EDA

- Understand dataset size and distribution across the 10 classes
- Check for class imbalance
- Visually inspect sample images from each class
- Verify image resolution, format, and color channels
- Identify corrupted, duplicate, or mislabeled images
- Analyze pixel intensity patterns across classes

- Determine preprocessing needs (normalization, resizing, augmentation)

Missing Value Analysis

To verify data completeness, a file-level check was performed across the dataset:

- All 10 class folders were confirmed to be present and non-empty
- No missing, empty, or unreadable image files were found
- No corrupted images that failed to load were detected during inspection
- Every class folder contained a consistent number of samples with no unexpected gaps

Image Size Analysis

The dimensions of all images were extracted and summarized. All images in the EuroSAT MS dataset are 64×64 pixels across all 13 spectral bands, with consistent dimensions throughout the dataset. Uniform image sizes simplify preprocessing since no resizing is required before feeding data into the models.

Key Findings

- The dataset contains 27,000 satellite images across ten land use and land cover classes
- Each image contains 13 spectral bands rather than just RGB, capturing additional information such as near-infrared and short-wave infrared reflectance
- No missing values exist, since labels are directly derived from folder structure
- The dataset is relatively balanced across classes, with each class containing between 2,000 and 3,000 images
- All images have consistent 64×64 pixel dimensions across all bands, making preprocessing straightforward
- Certain bands (e.g., near-infrared) show clearer separation between vegetation-related classes compared to RGB alone
- No corrupted or duplicate files were found during inspection

Conclusion

The Exploratory Data Analysis provided a comprehensive understanding of the EuroSAT multispectral dataset and identified several important characteristics relevant to land use and land cover classification. The analysis confirmed class balance, consistent image dimensions across all 13 bands, and clean data with no missing values or corruption. Using the multispectral version over RGB-only preserves additional spectral information useful for distinguishing between similar land cover types. These findings will guide the subsequent preprocessing, band selection, and model development stages.